

Zentra Platform Specification v1.0

Status: APPROVED FOR IMPLEMENTATION

Purpose: Source of truth for Cursor

Goal: Build a working MVP in 24 hours

1. Product Vision

Zentra is a multi-tenant appointment booking SaaS platform for appointment-based businesses.

It is not a marketplace.

It is not white-label.

It is one shared platform where many independent businesses operate inside the same app.

Appointment booking is the entry point. The long-term vision is to become the operating system for appointment-based businesses.

2. Supported Business Types

Zentra should support businesses such as:

- Barbers
- Hair salons
- Nail artists
- Dentists
- Therapists
- Coaches
- Clinics
- Personal trainers
- Consultants

Each business is independent.

3. Platform Model

The platform uses:

- One mobile app
- One backend
- One PostgreSQL database
- Shared schema
- Logical tenant isolation

No separate apps, databases, or backends per business.

4. Core Concept: Business / Tenant

A tenant means one independent business.

In product language, use `Business`.

In architecture, each business is a tenant.

Example:

- Jack Sparrow's Barber Shop
- Dr. Nicholas Dental Clinic
- Luxury Nails

Each business owns its own:

- services
- staff
- clients
- appointments
- settings

Data must never be shared across businesses unless explicitly approved in the future.

5. Users

Users are global identities.

One user account can belong to multiple businesses.

Example:

Ali can be:

- owner of Ahmad Barber
- staff in Luxury Nails
- client in Prime Therapy

The user account is global. Business access is controlled by memberships.

6. Roles

MVP roles:

- Owner
- Staff
- Client

Future roles:

- Admin
- Platform Admin

Do not implement Platform Admin now.

7. MVP Scope

Build the smallest working app that supports:

Authentication

- Register
- Login
- Logout
- Refresh token
- Authenticated API access

Business Management

- Create business
- View my businesses
- View business details
- Update business basic info

- Deactivate business

Memberships

- Owner membership when business is created
- User can belong to multiple businesses
- Membership connects user + business + role

Services

Each business can create its own services.

A service has:

- name
- description
- duration in minutes
- price
- active/inactive status

Service duration is custom per business.

Example:

Business A:

- Haircut: 30 minutes
- Beard: 10 minutes

Business B:

- Haircut: 45 minutes
- Beard: 15 minutes

No global shared service duration.

Staff

A business can have staff members.

A staff member belongs to one business.

For MVP, staff can be simple:

- user
- business
- display name

- active/inactive

Appointments

A client can book one appointment.

Appointment MVP rules:

- one business
- one service
- one staff member
- one client
- one start time
- one end time
- one status

Appointment statuses:

- BOOKED
- CANCELLED
- COMPLETED

No group appointments.

No multi-service appointments.

No multi-staff appointments.

No rooms.

No equipment.

8. Appointment History Rule

Appointments are historical records.

When an appointment is booked, store a snapshot of:

- service name at booking
- service duration at booking
- service price at booking
- staff name at booking
- appointment start time
- appointment end time

If a business later changes a service duration, name, price, or staff configuration, old appointments must not change.

9. Business Categories

Each business may belong to one category.

Examples:

- Barber
- Dentist
- Nail Artist
- Therapist
- Coach
- Clinic
- Personal Trainer
- Consultant

This is metadata only.

This does not make Zentra a marketplace.

For MVP, categories can be simple static values.

10. Tenant Isolation Rules

Every business-owned resource must include `businessId`.

Backend must always verify that the authenticated user has access to that business before allowing access.

Frontend filtering is not security.

Security must be enforced in the backend.

11. Mobile App MVP Screens

Build simple screens only.

Auth

- Register
- Login

Main

- My Businesses
- Create Business
- Business Dashboard

Business Dashboard

Show:

- business name
- category
- services
- staff
- appointments

Actions:

- manage services
- manage staff
- view appointments
- create appointment

Services

- list services
- create service
- update service
- deactivate service

Appointments

- list appointments
- create appointment
- cancel appointment
- complete appointment

UI should be clean, simple, and usable.

Do not spend time on advanced design.

12. Backend Stack

Use:

- NestJS
- TypeScript
- PostgreSQL
- Existing project structure
- Existing auth foundation

Use the ORM already present in the project. If ORM is not initialized yet, use the simplest consistent approach already used by the codebase.

Do not redesign the backend.

Do not introduce microservices.

13. Mobile Stack

Use:

- React Native
- Expo
- TypeScript

Keep UI simple.

Prioritize functionality over visual polish.

14. API Expectations

Create REST APIs for:

- businesses
- memberships
- services
- staff
- appointments

Use authenticated endpoints.

Use consistent response shapes.

Validate input on the backend.

Return clear error messages.

15. Out of Scope for 24-Hour MVP

Do not implement:

- payments
 - notifications
 - automatic rebooking
 - revenue dashboard
 - smart empty slot filling
 - platform admin
 - advanced analytics
 - advanced staff schedules
 - availability engine
 - recurring appointments
 - deposits
 - multi-location businesses
 - public marketplace
 - tenant switching polish
 - advanced permissions
 - branding/themes
 - AI features
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16. Future Roadmap Ideas

Do not implement now, but do not block future support for:

Automatic Rebooking

Suggest repeat appointments based on client behavior.

Revenue Dashboard

Show business owners revenue, appointment trends, repeat clients, and no-shows.

Smart Empty Slot Filling

Optional feature where a business can notify clients about open slots.

This must be opt-in per business.

17. Implementation Priorities

Cursor should implement in this order:

1. Confirm app runs.
 2. Confirm auth works.
 3. Add business entity/model.
 4. Add memberships.
 5. Add services.
 6. Add staff.
 7. Add appointments.
 8. Add basic mobile screens.
 9. Add validation.
 10. Add tests where practical.
 11. Run build/test.
 12. Provide final implementation summary.
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18. Definition of Done

The MVP is done when:

- user can register/login
 - user can create a business
 - user becomes owner of that business
 - user can view their businesses
 - user can create services for a business
 - user can create staff for a business
 - user can create appointments
 - appointments store historical service snapshot
 - business isolation is enforced in backend
 - mobile app can perform the core flows
 - backend builds successfully
 - mobile app starts successfully
 - tests pass where implemented
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19. Cursor Rules

Cursor must:

- implement only this specification
- avoid overengineering
- keep code simple
- avoid future-phase features
- stop when MVP is working
- report changed files
- report assumptions
- report unimplemented items
- report how to run the app

Cursor must not:

- redesign architecture
 - add marketplace behavior
 - add payments
 - add analytics
 - add notifications
 - add AI features
 - introduce unnecessary abstractions
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20. Final Goal

By the end of implementation, Zentra should be a working MVP where multiple businesses can exist in one platform, each business manages its own services, staff, and appointments, and data remains isolated per business.

The goal is not perfection.

The goal is a working, clean, extendable MVP.